

KARIM TIT

French national, 30 years old, 📍 Campus Kirchberg, University of Luxembourg

Postdoctoral Researcher in Machine Learning

✉ karim.tit@uni.lu —  karim-tit —  karimtito —  karimtito.github.io —  Scholar profile

WORK EXPERIENCE 🏢

University of Luxembourg

May 2024 - Ongoing

Postdoctoral Researcher, supervised by Dr. Maxime Cordy

Working on formal methods and statistical model-checking methods for Deep Learning certification.

Thales Land and Air Systems (LAS)

February 2021 - January 2024

ML Research Engineer, supervised by Dr. Louis-Marie Traonouez

Worked on the reliability of Deep Neural Networks for cybersecurity applications.

KeyQuant

May 2020 - September 2020

Research internship, supervised by Dr. Charles Chevalier

Enhanced and implemented [in Python] trend-following trading strategies with machine learning models.

Capital Fund Management & École Polytechnique

March 2019 - July 2019

Quantitative research intern, supervised by Dr. Michael Benzaquen & Dr. Iacopo Mastromatteo

Developed a financial market microstructure model for the Chair Econophysics and Complex Systems

EDUCATION 🎓

INRIA & University of Rennes

February 2021- April 2024

Ph.D. in Machine Learning, supervised by Dr. Teddy Furon & Dr. Mathias Rousset

Thesis subject : "Reliability of Deep Neural Networks with Rare Event Simulation Algorithms."

ENSAE Paris, Polytechnic Institute Of Paris

2018-2020

Master of Engineering and Data Science (Diplôme d'ingénieur)

highest distinction, GPA 4.0

Paris-Saclay University

2019-2020

Masters in Mathematics of Randomness (M2), Statistical Learning track

highest distinction, GPA 4.0

Paris Sciences Et Lettres University

2018-2019

Research Masters in Pure and Applied Mathematics (M2)

highest distinction , GPA 4.0

Sorbonne University

2016-2018

M.Sc. in Mathematics 1st year (M1),

highest distinction, GPA 4.0

B.Sc. in Mathematics with minor in Physics (L3),

highest distinction, GPA 4.0

LANGUAGE & TECHNICAL SKILLS ✂

French

Mother tongue

English

Fluent (C1 level, TOEFL : 104/120)

Programming languages

Python 🐍 (main), C++, Java (secondary)

Python libraries

Numpy, Pandas, PyTorch, TensorFlow, Scikit-learn

Operating systems

🐧, 🖥

Software tools

git, 📁, L^AT_EX

TEACHING 📖

University of Luxembourg

September 2024- Ongoing (Yearly)

Classes on Robust Machine Learning for Data Science & Cyber-Security master's students (12×2 h).

Classes on Software Engineering for the Computer Science Bachelor students (9h).

École Polytechnique

October 2019- January 2020

Tutorials for undergrad bachelor students covering advanced statistics and measure theory (45h).

Stay Within Your Bounds: Distance-Guided Decoding for Guaranteed Context-Free Grammar Compliance

V. Collura*, K. Tit*, E. Giunchiglia, M. Papadakis, M. Cordy - Conference on Empirical Methods in Natural Language Processing (EMNLP), Findings of the Association for Computational Linguistics, Oct 2026, Budapest, Hungary.

*Equal contribution.

DivKC: A Divide-and-Conquer Approach to Knowledge Compilation

O. Zeyen, K. Tit, M. Cordy, G. Perrouin - 29th International Conference on Fundamental Approaches to Software Engineering (FASE), April 2026, Turin, Italy.

On the Robustness of Tabular Foundation Models: Test-Time Attacks and In-Context Defenses

M. Djilani, T. Simonetto, K. Tit, F. Tambon, S. Ghamizi, M. Cordy, M. Papadakis - 4th IEEE Conference on Secure and Trustworthy Machine Learning (SatML), March 2026, Munich, Germany.

KCLNet: Physics-Informed Power Flow Prediction via Constraints Projections

P. Dogoulis, K. Tit, M. Cordy - European Conference in Machine Learning (ECML), Aug 2025, Porto, Portugal.

TS-SCAN: a Density-Based Clustering Method for Trajectory Analysis and Anomaly Detection.

L. Monnier, LM. Traonouez, K. Tit - 2024 International Radar Conference (RADAR), Nov 2024, Rennes, France.

Fast Neural Networks Reliability Estimation with Attack-Driven Importance Sampling.

K. Tit, T. Furon - 40th Conference on Uncertainty in Artificial Intelligence (UAI), Jul 2024, Barcelona, Spain.

Three Bricks to Consolidate Watermarks for Large Language Models.

P. Fernandez, A. Chaffin, K. Tit, V. Chappelier, T. Furon - IEEE 15th International Workshop on Information Forensics and Security (WIFS), Dec 2023, Nuremberg, Germany.

Gradient-Informed Neural Network Statistical Robustness Estimation.

K. Tit, T. Furon, M. Rousset - 26th International Conference on Artificial Intelligence and Statistics (AISTATS), Apr 2023, Valencia, Spain.

Efficient Statistical Assessment of Neural Network Corruption Robustness.

K. Tit, T. Furon, M. Rousset - 35th Conference on Neural Information Processing Systems (NeurIPS), Dec 2021, Sydney (virtual), Australia.

Évaluation statistique efficace de la robustesse de classifieurs (*In French*).

K. Tit, T. Furon, M. Rousset, L.-M. Traonouez - Conference on Artificial Intelligence for Defense (CAID), DGNUM, Nov 2021, Rennes, France.

ACADEMIC AWARDS

- Top Reviewer Award (top 10%) at Neurips 2025.
- Best Student Paper Award at WIFS 2023, jointly with co-authors P. Fernandez and A. Chaffin.
- Scholarship for academic excellence in Mathematics, 'Sophie Germain' 2019-2020
Masters Scholarship awarded by the Fondation Mathématiques Jacques Hadamard (FMJH)
- Scholarship for academic excellence in Mathematics, 'PGSM' 2018-2019
Masters Scholarship awarded by the Paris Graduate School of Mathematics